

# JAVIER ROMÁN PÁSARO

## AI / PYTHON ENGINEER – NLP & SPEECH PROCESSING

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[LinkedIn](#) | [Google Scholar](#)

Engineer specialized in Artificial Intelligence and Python backend development. Experience in automatic speech transcription systems and natural language processing using speech recognition models and real-time web technologies. Interested in developing backend applications that integrate AI models and data processing pipelines.

## TECHNOLOGIES

Python · FastAPI · SQLAlchemy · MySQL · Machine Learning · Whisper / Speech Recognition · Transformers · NLP · REST APIs · WebSockets · WebRTC · Linux · Git · HTML · CSS · JavaScript

## PROFESSIONAL EXPERIENCE

### AI Backend Developer – SCRIBAL (Universitat de Barcelona)

Feb. 2025 – Jun. 2026

Development of a web platform for multilingual real-time audio transcription and translation, designed for educational environments.

- Backend development in Python (FastAPI) for a multilingual transcription and translation platform.
- Implementation of backend APIs for session management, users, and audio processing.
- Integration of Whisper-based ASR models for automatic speech recognition.
- Real-time audio processing using WebRTC and WebSockets.
- Pipeline for transcription of audio/video files with export to SRT and VTT formats.
- Database design using SQLAlchemy and MySQL for sessions and user management.
- Deployment and maintenance of the platform in cloud environments using virtual machines, storage, and remote databases.

### NLP Research Engineer - NON-CONSPIRA-HATE (University of Huelva)

Nov. 2023 – Dec. 2024

- Development of Python scripts for dataset preparation and NLP experiments.
- Training and evaluation of Transformer and LLM-based models for hate speech detection.
- Development of a demonstration web application for tweet classification.
- Configuration of deep learning environments on Linux with GPU support (CUDA, PyTorch, TensorFlow).

## EDUCATION

- **Master's Degree in Computer Engineering – Big Data & Cloud Computing**  
University of Huelva | 2023 – 2026
- **Bachelor's Degree in Computer Science (Erasmus Programme)**  
University of Silesia, Katowice | 2021 – 2022
- **Bachelor's Degree in Computer Engineering – Mention in Computer Science**  
University of Huelva | 2019 – 2023

## PUBLICATIONS

- **SCRIBAL**: A Digital Transcription Tool in Higher Education
- **I2C-UHU at IberLef-2023 HOMO-MEX** task: Ensembling Transformers models to Identify and Classify Hate Messages Towards the LGBTQ+ Community
- **I2C-UHU at IberLEF-2024 HOMO-MEX** task: Leveraging Large Language Models and Ensembling Transformers to Identify and Classify Hate Messages Towards the LGBTQ+ Community
- **I2C-UHU at EXIST 2024**: Transformer-Based Detection of Sexism and Source Intention in Memes Using a Learning with Disagreement Approach

## LANGUAGES

- **Spanish** – Native
- **English** – B2 (Cambridge First Certificate)
- **French** – B1 (DELF)

## CERTIFICATIONS & CONFERENCES

- **CCNA1** - Cisco Routing & Switching
- XL **SEPLN** Conference 2024 (IberLEF)
- **INTERSPEECH** 2025